

LABORATORY RECORD

Course: Full Stack Web Development
Course Code: 23CM4121
Experiment No.: 4
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1. TITLE

Design and Implementation of Dynamic Web Application Using Advanced DOM Operations, Timers, and Theme Switcher

2. AIM

To develop a dynamic single-page web application featuring interactive student greeting generation, a real-time digital clock using JavaScript interval timers, dynamic task list management, and customizable theme switching via inline style manipulation.

3. OBJECTIVE

The objectives of this experiment are:

- To manipulate DOM text nodes dynamically using `textContent` and user input values.
 - To implement asynchronous time-based execution using `setInterval()` for a live digital clock.
 - To programmatically construct and append dynamic list items (`document.createElement('li')`) to a target container.
 - To bind HTML button event attributes (`onclick`) directly to JavaScript event handler functions.
 - To dynamically update document styling properties (`document.body.style`) for theme customization.
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4. THEORY

Dynamic web applications rely on JavaScript Document Object Model (DOM) APIs to modify page structure and styling at runtime. The `document.getElementById()` method retrieves DOM nodes, while `textContent` updates text content safely without rendering raw HTML.

Asynchronous timers such as `setInterval(callback, delay)` repeatedly execute functions at specified millisecond intervals, making them ideal for real-time components like digital clocks (`Date().toLocaleTimeString()`).

DOM manipulation functions like `document.createElement('li')` and `appendChild()` dynamically add nodes to the document tree. Direct style property mutation (`document.body.style.background, color`) enables immediate visual theme changes across light, dark, and custom color themes.

Application Flow:

HTML5 UI Component Layout -> User Input Capture & Greeting -> Real-Time Clock Interval Loop -> Dynamic Task Node Appending -> Dynamic Theme Style Mutation

5. ALGORITHM / PROCEDURE

1. Create an HTML5 document structure with header 'Dynamic Web Application' and 4 distinct feature box containers (.box).
 2. Style box containers with white background, rounded corners (8px), padding (20px), and box-shadows.
 3. Implement `greetUser()` function to read `#userName` value and update `#greetMsg` with welcome greeting text.
 4. Implement `showTime()` function to retrieve current time via `new Date().toLocaleTimeString()` and update `#clock`.
 5. Set an interval timer using `setInterval(showTime, 1000)` to update clock display every second.
 6. Implement `addNewTask()` function to read `#taskInput`, trim whitespace, create new `<li>` node, append to `#taskContainer`, and clear input.
 7. Implement `setTheme(bg, color)` function to update `document.body.style.background` and `color` dynamically.
 8. Bind buttons to `onclick` attributes: `greetUser()`, `addNewTask()`, `setTheme('#f4f7fb', '#222')`, `setTheme('#0f172a', '#f8f9fa')`, `setTheme('#ecfdf5', '#064e3b')`.
 9. Open web page in browser, test greeting, verify digital clock seconds counter, add tasks, and test theme switcher buttons.
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6. PROGRAM

index.html

```
<!DOCTYPE html>
<html>

<head>
  <title>Dynamic Web Application</title>
  <style>
    * {
      margin: 0;
      padding: 0;
      box-sizing: border-box;
    }

    body {
      font-family: Arial, sans-serif;
      background: #f4f7fb;
      padding: 20px;
      color: #222;
    }

    header {
      background: #1e293b;
      color: white;
      padding: 15px;
      text-align: center;
      border-radius: 8px;
    }

    .box {
      background: white;
      padding: 20px;
      margin: 15px 0;
      border-radius: 8px;
      box-shadow: 0 2px 8px rgba(0, 0, 0, 0.08);
    }

    input {
      padding: 8px;
      width: 60%;
      margin-right: 8px;
    }

    button {
      padding: 8px 14px;
      background: #2563eb;
      color: white;
      border: none;
      border-radius: 4px;
      cursor: pointer;
    }

    button:hover {
      background: #1d4ed8;
    }

    #clock {
      font-size: 26px;
      font-weight: bold;
      margin-top: 10px;
      color: #2563eb;
    }

    li {
      margin: 6px 0;
      padding: 6px;
      background: #f1f5f9;
      border-radius: 4px;
    }
  </style>
</head>

<body>
```

```
<header>
  <h1>Dynamic Web Application</h1>
  <p>Interactive DOM Operations</p>
</header>

<div class="box">
  <h3>Student Greeting</h3>
  <input id="userName" placeholder="Enter your name">
  <button onclick="greetUser()">Greet</button>
  <p id="greetMsg" style="margin-top: 10px; font-weight: bold;"></p>
</div>

<div class="box">
  <h3>Digital Clock</h3>
  <div id="clock">00:00:00</div>
</div>

<div class="box">
  <h3>Task List</h3>
  <input id="taskInput" placeholder="Add new task">
  <button onclick="addNewTask()">Add Task</button>
  <ul id="taskContainer" style="margin-top: 10px; list-style: none;"></ul>
</div>

<div class="box">
  <h3>Theme Switcher</h3>
  <button onclick="setTheme('#f4f7fb', '#222')">Light</button>
  <button onclick="setTheme('#0f172a', '#f8faff')">Dark</button>
  <button onclick="setTheme('#ecfdf5', '#064e3b')">Emerald</button>
</div>

<script>
  function greetUser() {
    const name = document.getElementById('userName').value;
    document.getElementById('greetMsg').textContent = name ? "Hello, " + name + "! Welcome to ANITS." : "";
  }

  function showTime() {
    document.getElementById('clock').textContent = new Date().toLocaleTimeString();
  }
  setInterval(showTime, 1000);
  showTime();

  function addNewTask() {
    const val = document.getElementById('taskInput').value.trim();
    if (val) {
      const li = document.createElement('li');
      li.textContent = val;
      document.getElementById('taskContainer').appendChild(li);
      document.getElementById('taskInput').value = '';
    }
  }

  function setTheme(bg, color) {
    document.body.style.background = bg;
    document.body.style.color = color;
  }
</script>
</body>
</html>
```

7. CODE EXPLANATION

Code / Syntax	Explanation
<code>document.getElementById('userName').value</code>	Extracts user-entered string value from input element.
<code>document.getElementById('greetMsg').textContent</code>	Dynamically updates text node content of greeting paragraph.
<code>new Date().toLocaleTimeString()</code>	Formats current time into locale-specific string representation.
<code>setInterval(showTime, 1000)</code>	Invokes showTime callback function repeatedly every 1000 milliseconds.
<code>document.createElement('li')</code>	Programmatically instantiates new HTML list item element node.
<code>document.getElementById('taskContainer').appendChild(li)</code>	Appends newly created list item node to target element.
<code>document.body.style.background = bg</code>	Dynamically mutates body element background style property for theme switching.

8. TEST CASES AND EXECUTION DATA

The experiment was executed with the following test cases to verify functionality:

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Page Initial Load	Open webpage in browser	Header, 4 feature boxes, and initial clock 00:00:00 display	Pass
TC-02	Greet User Function	Enter 'raghu ram' & click Greet	Displays 'Hello, raghu ram! Welcome to ANITS.'	Pass
TC-03	Digital Clock Interval	Observe clock display	Clock updates every second with live local time string	Pass
TC-04	Add Task Node	Enter 'Fs' & click Add Task	New list item 'Fs' appended dynamically to Task List container	Pass
TC-05	Theme Switcher	Click Light / Dark / Emerald	Body background and text colors update instantly to theme palette	Pass

9. OUTPUT

Output 1: Dynamic Web Application Dashboard (Greeting, Clock, Task List & Theme Switcher)

The screenshot displays a web application dashboard with a dark blue header titled "Dynamic Web Application" and a subtitle "Interactive DOM Operations". The dashboard is divided into four main sections:

- Student Greeting:** A text input field containing "raghu ram" and a blue "Greet" button. Below the input, the text "Hello, raghu ram! Welcome to ANITS." is displayed.
- Digital Clock:** A section showing the current time as "6:17:21 PM" in blue text.
- Task List:** A text input field with the placeholder "Add new task" and a blue "Add Task" button. Below the input, a single task "Fs" is listed.
- Theme Switcher:** A section with three buttons: "Light", "Dark", and "Ethereal". The "Dark" button is currently selected.

10. RESULT

Thus, the dynamic web application incorporating interactive greeting generation, real-time digital clock, dynamic task node creation, and body theme switcher was successfully designed, implemented, and verified.